STOCHASTIC METHODS IN WATER RESOURCES

Unit 2: Hydrological statistics and extremes
Lecture 8: Hydrological design, Makov chains and regional frequency analysis

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Hydrologic design

Generalities

- Planning and management of water resources can be classified according to their 1) **use** (e.g. Hydroelectricity, water supply, recreation, etc) and their 2) **control** (e.g. drainage, flood, contamination, sedimentation, etc.).
- The **hydrologic design** for control is relate to extreme events of short duration (e.g. a flood). The optimal design is a balance between the cost and safety of the designed structure.
- Regarding that water is finite on earth, extreme hydrological events have limit values that are difficult to know and thus are quite uncertain. For precipitation data, the superior limit is defined by the probable maximum precipitation (PMP) and for streamflow data it is defined by the probable maximum flood (PMF). This limits are used for instance in the design of excess spillways in reservoir.
- The estimation of the limits is deterministic so that the limits do not obey to a return period. An structure designed based on those limits might support all possible of hazards.

Generalities

- When the limits are not considered, the hydrologic design of the structure is based on frequency analsis, where the magnitude of the hydrological variable of interest is estimated for certain design return period (T_D). For instance, for $T_D = 100$ yr the corresponding value of the design hydrological variable is expected to be exceeded in average once every 100 yr, or that the probability of the occurrence of that event in a given year is $1/T_D = 0.01$, which is quite low.
- Note that when a structure is design for a T_D, there is a residual hazard because there is a probability that a larger event occur during the structure's design life.
- To estimate the magnitudes of the design hydrological variables is needed to obtain the frequency curves; a methodology that has been explained before here. Note that the frequency curves are obtained assuming that the data sample (e.g. annual time series of extreme rainfall) comes from a stationary process.

Hydrologic design

Definition of design return period (T_D)

- A design of a structure can be based on the **design return period (T_D)**. Their definition is based on design manuals or design codes (e.g. manual de diseño del INVIAS o de la EAAB). Also, the T_D can be established after risk analysis in hydrology or from economical analysis, these two allow estimate an optimal T_D.
- The magnitude to the hydrological design variable is obtained from the frequency curve using T_D. However, it is important to follow the recommendation given by design manuals and codes regarding the type of probability distributions to be implemented and the methods to estimate their parameters. For instance, in USA agencies is recommended to use the Log Pearson type III for the analysis of maximum streamflows.

How to determine T_D?

- According to the type of structure, and based on previous engineering experience and the recommendations given by design manuals and codes, a value or a range of values for T_D is given. For instance, for the design of highway drainage systems it is recommended $5yrs \leq T_D \leq 100yrs$, where the value depends on the traffic volume. Similarly, while a $2yrs \leq T_D \leq 25yrs$ is recommended for pluvial sewage systems in small cities, a $25yrs \leq T_D \leq 50yrs$ is recommended for large cities.
- Based on government policies, the level of hazard can dictate the range of T_D. So for instance, in coastal cities, the risk of floods can be higher so for the design of coastal levees, T_D can be up to 10000 yrs.

Definition of design return period (T_D)

How to determine T_D?

- In Colombia, the **RAS 2000** technical regulations, stablish that a $2yrs \leq T_D \leq 50yrs$ for the design of combine or pluvial sewage systems. The ultimate value of T_D will depend on the use, slope, area, etc of the system. The **INVIAS** stablish the following T_D for the design of highways structures:
  - Gutters: $T_D = 5yrs$
  - Culvert $\phi = 0.9$ m: $T_D = 10yrs$
  - Culvert $\phi > 0.9$ m: $T_D = 20yrs$
  - Bridge $10m \leq$ span $\leq 50m$: $T_D = 50yrs$
  - Bridge span $> 50m$: $T_D = 100yrs$

Selection of T_D based on hydrological risk

To estimate T_D is assumed a value for the **hydrological risk (R_h)**. Recall that R_h is defined as the probability that the hydraulic capacity of the structure is exceeded by a hydrological hazard once during the expected life of the structure (n yrs). Accordingly

$$T_D = \frac{1}{1 - (1 - R_h)^{1/n}}$$

Definition of design return period (T_D)

Selection of T_D based on hydrological risk

- The table shows the values of T_D for different values of R_h and n. Note that the lesser R_h the higher T_D. In contrast, the larger n the highter T_D. Remember that when $T_D = n$, $R_h \approx 0.63$, which is quite high.
- For instance, in the design of a cofferdam and a diversion tunnel, it is common to adopt a streamflow design for a $T_D = 50$yrs, which correspond to $R_h = 0.096$ or 9.6% for $n = 5$yrs. This contrast, T_D for a design seismic earthquake, according to the design codes in Colombia, is equal to 475 yrs, which correspond to $R_h = 0.1$ and $n = 50$yrs.

R_h	Vida útil esperada n (años)					
	5	10	25	50	75	100
0,01	498	995	2488	4975	7463	9950
0,05	98	195	488	975	1463	1950
0,1	48	95	238	475	712	950
0,2	23	45	113	225	337	449
0,3	15	29	71	141	211	281
0,4	10	20	49	98	147	196
0,5	8	15	37	73	109	145
0,6	6	11	28	55	82	110
0,7	5	9	21	42	63	84
0,8	4	7	16	32	47	63
0,9	3	5	11	22	33	44
0,95	2	4	9	17	26	34

Conversely,

$$R_h = 1 - \left(1 - \frac{1}{T_D}\right)^n$$

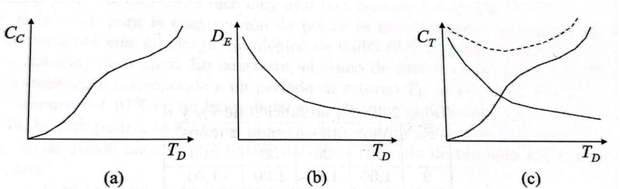
Note that when T_D is low, R_h is total (absolute certainty of the occurrence of the event during n). For instance for the case of a pluvial sewage system, when $T_D = 2$ yrs and $n = 25$yrs, $R_h \approx 1$. This means that with total certain the sewage system will experiment at least once an event that exceed its hydraulic capacity.There are other uncertainty associated to the structural design, construction, operation, etc.

Definition of design return period (T_D)

Selection of T_D based on economical analysis

The effects of the occurrence of extreme hydrological events on the population and economy change depending on the event and population characteristics. To establish an optimal T_D is necessary to perform an economical analysis. There are two types: **the analysis of minimal cost** and **the analysis of cost-benefits**.

- Analysis of minimal cost: The higher T_D the higher the cost of the control measures (capital costs (C_C)) adopted to reduce the risk (e.g. the construction of a most robust structure). The following figure shows



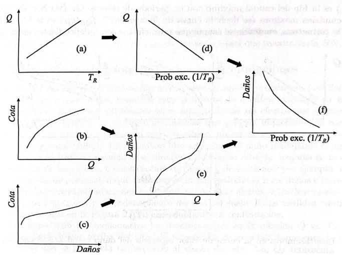
  - The figure a, C_C vs T_D, shows that the annual capital cost Cc increase when T_D increase.
  - Given a T_D, there is a residual hazard, which represents the possible inevitable damages or risks. So, the figure b represent the relations of the expected value of the damage D_E due to a residual hazard and the T_D. Note that D_E increase when T_D decrease (more residual hazard).
  - The total cost C_T is thus $C_T = C_C + D_E$. The curve in figure c shows C_T vs T_D (dotted line), from where the optimal T_D is the one that minimize C_T. This means that

$$\frac{dC_T}{dT_D} + \frac{dC_C}{dT_D} + \frac{dD_E}{dT_D} = 0$$

Definition of design return period (T_D)

One of the problems is how to make the curve D_E vs T_D (see figure b above). To make it, follows this procedure:

- Suppose that a flood control structure is going to be constructed.
- From hydrological studies, is possible to establish the frequency curve (T_R vs Q) (see figure a below) and the calibration curve of one of the sections of the river (water level (wl) vs Q) (see figure b below).
- With formation regarding areas prone to be flooded, it is possible to build the damage curve (water level (wl) vs damages) (see figure c below).
- The frequency curve (see figure a below) can be transformed into a relations of Q vs the exceedance probability ($1/T_R$) (see figure d below).
- Based on the curves in figure b and y is possible to make the curve Q vs damages (see figure e)
- Finally, based on curves in figure d and e is possible to construct the curve of exceedance probability vs damages (see figure f).

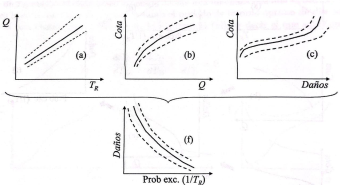


Definition of design return period (T_D)

The curve of exceedance probability vs damages shows the resulting damage if a flood with a T_R occurs. If the design streamflow (Q_D) is associated to a T_D, the structure would be able to prevent those damages caused by $Q \leq Q_D$, but not for $Q > Q_D$. This means that the expected value of the damage D_E in a given year is

$$D_E(T_D) = \int_{Q_D}^{\infty} D(q) f_Q(q) dq$$

where $D(q)$ is the associated damage to the occurrence of $Q = q$ and $f_Q(q)$ is the pdf of the maximum streamflows. Note that $1/T_R$ is equivalent to $1 - F_Q(q)$, where $F_Q(q)$ is the cdf of Q. It is important to mention that $f_Q(q)$ is well known, so Q_{T_R} is certain. This equation represent the are under the curve between 0 and $1/T_D = 1/T_R$. The curve of D_E vs T_D is built using the equation above for different values of T_D. The uncertainties associated to the different curves used to build D_E vs T_D are shown in the figure below.



Definition of design return period (T_D)

- Analysis of cost-benefits: this analysis requires to consider all costs and benefits of the risk management policies in order to:
  1. stablish the risk scenario (how large are the damages per year)
  2. determine the potential risk mitigation measures and their costs
  3. compare the benefits and the costs of reducing risk (are the benefits larger than the costs?)
  4. compare the relationships between costs and benefits for the different measures (which bring larger net benefits?)

  The Analysis of cost-benefits for flood events is made based on the flood type and its characteristics (e.g. water depth, duration, flow speed, rate of water rise, frequency, contamination, etc.). The analysis try to quantify the variety of damages and their associated costs, during and after the occurrence of the extreme event. According to the USA Army Corsps of Engineers, the benefits are clasified as:
  1. Benefits due to flood mitigation (B_{IR}) that reduce economic losses.
  2. Benefits due to increments (B_I) of activities due to implemented measures.
  3. Benefits due to increments of incomes (B_L) due to the implemented measures (e.g. can bring more development to the area).

  Accordingly, the economic efficiency due to the implementation of one or multiple mitigation measures for flood events can be estimated as a function of T_D as

$$BN(T_D) = B_{IR}(T_D) + B_I(T_D) + B_L(T_D) + C(T_D)$$

  where BN is the net benefit and C represents the total cost to implement, operate, and maintain the adopted measure(s).
  $B_{IR}(T_D) = D_E(T_D)|_{without} - D_E(T_D)|_{with}$, this is the difference between the damages without the control measures (current conditions) and the damages with the control measures (estimated using the equation for $D_E(T_D)$).

Measures to mitigate flood damage

Floods are perhaps the most common and frequent extreme hydrological events. Some measures to mitigate their effects are presented here that might help in the context of hydrologic design. These measures are classified as:

1. **structural**: these seeks to reduce damages by controlling the water flowing into the floodplain using reservoirs to store and delay the inflowing water volumes, increasing the capacity of channels to transport flow volumes, and building dikes and diversions.
2. **non structural**: Community preventing actions oriented towards reducing the susceptibility to damages due to floods, implementation of early warning systems to enhance preparedness and regulation policies related to land usages and relocation.

Measures to mitigate flood damage

Some of these structural and non structural measures modify the hydraulic and hydrologic relationships shown in a previous figure. Such modifications are explained below.

- **Reservoirs**: As showed in figure a, the presence of a reservoir upstream of the area that might be potentially flooded modify the Q vs T_R relationship (see the dotted line). So, for instance if Q_{0_e} is the control discharge of the reservoir (the outflow that is regulated or released from a reservoir) and T_{R_0} its return period, which can mean the maximum Q eager to transport the channel or that guarantee no damages downstream. Note that $Q < Q_{0_e}$ are regulated by the reservoir. However, this Q_{0_e} up to a return period T_{R_1}, but for $T_R > T_{R_1}$ the reservoir attenuation capacity decrease progressively and to the original Q vs T_R curve. The effects of the reservoir on the temporary storage of water are also shown in figure b.
- **Diversions**: These structures derive a portion of the flow volume transported by the main channel to reduce the volume that travel toward the area of interest. Figure c shows the effects of a diversion on the Q vs T_R curve (see dotted line). Again, analysing Q_0, in this case, Q_{0_d}, the there is not diversion if $Q < Q_{0_d}$. However, when $Q > Q_{0_d}$ the excess Q is diverted limiting the channel to transport Q_{0_d}. When Q increase the T_{R_1} increase and the capacity of the diversion tends to the maximum capacity, so that, the Q in the main channel will increase above Q_{0_d}.

Measures to mitigate flood damage

- **Hydraulic modifications of the channel**: The elevations above the channel banks can be reduced if the channel is modify (e.g. dredging and horizontal alignment modification) in order to increase its capacity. These modifications change the water level vs Q curve (see dotted line in figure d). These increments of the channel capacity upstream of the area of interest increase the water volume downstream affecting the area negatively, and modify Q vs T_R (see dotted line in figure e).
- **Levees**: are used to mitigate flood damage by preventing flooding. Levees modify the water level vs damage curve as is shown in the dotted line of figure f. Note that C_1 is the water level from which damages in the protected area occur with T_D. Also, levees change the water level vs Q curve (see dotted line in figure g), because the channel section can convey larger Qs. Similarly to the hydraulic modifications, levees trigger increases of the Qs transported downstream changing the Q vs T_R curve (see dotted line in figure h.
- **Individual measures against flooding**: These measures include the use of protective barriers and low walls to stop water from entering buildings that often get flooded. These measures modify the water level vs damage curve (see dotted line in figure f).
- **Relocation**: It consists of locating exposed and vulnerable structures in areas less susceptible to damage. Note that this modifies the water level vs damage curve (see dotted line in figure i), which tend to be vertical in case most of the structures are relocated. Relocation involves very high costs so its feasibility could be low.
- **Land-use planning**: Seek to reduce future vulnerability. It consist in involving building codes, relocation, and land-use change, etc.

Measures to mitigate flood damage